

LEC 7 - The Moon & Eclipses

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Moon Learning Goals

1. Sketch a diagram of the motion of the moon around the earth ☐
2. Predict the moons appearance and location based on its position in its orbit ☐
3. Explain the phenomena of lunar and solar eclipses based on the geometry of the Earth, Moon, and Sun ☐

Scale Model of Earth & Moon

Earths diameter $\rightarrow 12,740\text{km}$
 Moons diameter $\rightarrow 3,470\text{km}$ $\sim 0.25x$
 Distance between $\rightarrow 384,400\text{km}$ $\sim 30x$

The moons orbit around earth follows the same right-hand rule as the Earth's rotation and orbit

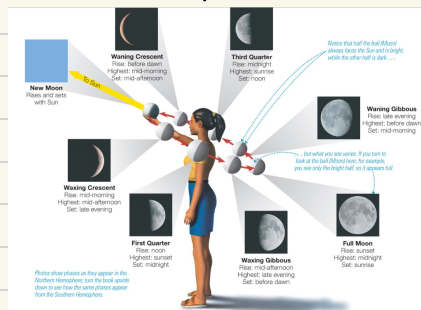
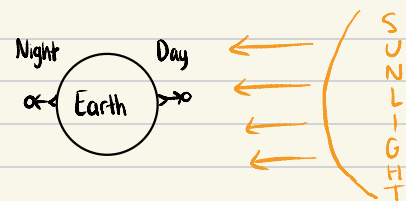
* Phases of the Moon are caused by the Moon's orbit around the earth and illumination by the Sun.

* Waning means the lit up portion of the moon is thinning, and waxing is when it is growing

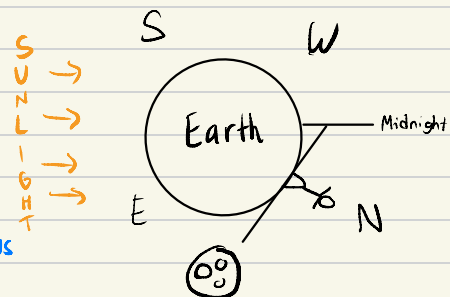
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- Rise/Set times depend on the Moons position in its orbit
 - $\rightarrow$ Objects rise and set over the horizon because of Earth's rotation
 - $\rightarrow$ Daytime: When your little slice of the Earth is pointed **towards** the Sun
 - $\rightarrow$ Nighttime: When your little slice of the Earth is pointed **away** from the Sun
- The phase of the Moon does **NOT** depend on the observer's location
 - $\rightarrow$ Only the appearance of the phase changes with observer's location



Suppose you go outside in Mississauga and look up at the sky at 10pm. The moon is visible towards the **east**, fairly low along the horizon. What phase is the moon in? **Waning Gibbous**



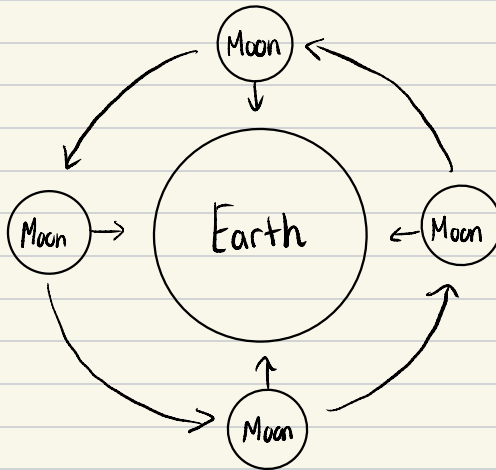
The Moon orbits the earth following the right hand rule, as the moon orbits Earth, the Sun illuminates different parts of the Moon, causing Moon phases

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Synchronous Rotation

- The Moon's rotation period is the same as its orbital period around the Earth
 - As a result, we always see the same half of the moon (near side)
- Note: Near side & Far side $\neq$ Light side & Dark side



* Eclipses were not covered this lecture *

◦ The same part of the Moon always faces the earth, causing a near & far side of the moon